Case 14: Two-Dimensional Spatial-Resolution Sensitivity

1 Verification purpose

This case is a negative verification test for mesh dependence in the current multiple-discrete-cell treatment of structures. A physical 10 m-by-10 m house may occupy one raster cell, several independent cells, one oversized cell, or disappear under categorical resampling as the mesh changes. Because those cells are not assembled into one conserved structure object, changing resolution alters represented heat release, firebrand generation, deposition, and structure-to-structure ignition.

The test asks whether all 15 input-matched simulations produce an evaluable community rate of spread (ROS), and whether the fitted values remain within the predeclared mesh-independence limits relative to the run-derived 10-m anchor. Detecting the known resolution sensitivity is useful diagnostic evidence, but it is a *verification failure*, not a pass.

2 Mathematical formulation and numerical implementation

2.1 Mode-resampled square array

The physical layout is a periodic array of square structures of side $a = 10\,\mathrm{m}$, separated in both coordinates by $d = 10\,\mathrm{m}$. Define the one-dimensional structural indicator

$$I(s) = \begin{cases} 1, & s \bmod (a+d) < a, \\ 0, & \text{otherwise.} \end{cases}$$

For grid interval $C_i = [i\Delta, (i+1)\Delta]$, preprocessing calculates

$$f_i = \frac{1}{\Delta}\int_{C_i} I(s)\,ds, \qquad b_i = \mathbf{1}(f_i \geq 1/2), \qquad B_{ij} = b_i b_j.$$

This coordinate-wise categorical-mode construction reproduces the reported geometries. At $\Delta = 7\,\mathrm{m}$, effective side lengths are 7 or 14 m and footprints range from 49 to $196\,\mathrm{m}^2$. At $\Delta = 10\,\mathrm{m}$, footprint and gap are exact. At $\Delta = 30\,\mathrm{m}$, one represented structure cell has area $900\,\mathrm{m}^2$ and represented structures are separated by 30 m. The 20- and 40-m maps eliminate represented firebreaks; this is an intended resampling diagnostic, not a preprocessing failure.

2.2 Cell-local firebrand treatment

A represented structural cell uses a design-fire curve $q''(t)$. The curve has a 300-s linear growth phase, a one-hour plateau at $400\,\mathrm{kW\,m^{-2}}$, and a 300-s decay. Under the current multiple-discrete-cell treatment,

$$\mathrm{HRR}_{ij}(t) = q''(t)\Delta^2, \qquad \dot{N}_{ij}(t) = GR\,\frac{\mathrm{HRR}_{ij}(t)}{1000}, \qquad GR = 10\,\mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

Changing Δ therefore changes both represented geometry and the heat-release rate (HRR) of each independent emitter. Firebrands use the structural empirical flight-distance distribution, Eulerian accumulation, and physical ignition. Non-structure cells use nonburnable fuel model 93, so vegetative surface spread is absent and every downwind ignition must be caused by deposited firebrands. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. The

current source independently refreshes initialized model-14 building transient HRR under this setting. The configured time-step scaling is documented by Qin et al. [2]:

$$\Delta t = \mathrm{CFL}\frac{\Delta}{U_{\mathrm{wind}}}, \qquad \mathrm{CFL} = 1, \qquad U_{\mathrm{wind}} = 17.9\,\mathrm{m\,s^{-1}}\ (40\,\mathrm{mph}).$$

The fractional step is required to hold this spatial sweep at fixed wind Courant–Friedrichs–Lewy (CFL). Preprocessing sets each generated stop to $\lceil 150000/\Delta t \rceil \Delta t$, then records the requested stop, aligned stop, and integer step count in the simulation specification.

For each represented downwind structure column s, postprocessing takes the median positive finite arrival time t_s. At least three ignited columns are required. Mean community ROS is the least-squares slope

$$ROS_\Delta = \frac{\sum_s (t_s - \bar{t})(x_s - \bar{x})}{\sum_s (t_s - \bar{t})^2}.$$

Because the 10-m grid exactly resolves both physical length scales, its fitted ELMFIRE result is the comparison anchor,

$$ROS_{10} = ROS_{\Delta=10\,\mathrm{m}}.$$

The dashed reference line and the 10-m ELMFIRE point therefore use the same calculated value; no rounded external estimate is substituted.

3 Derivation of expected behavior

A resolution-independent structure treatment would conserve each house's physical footprint, integrated heat release, firebrand source, accumulated load, and ignition state. Subject to numerical error, it would give $ROS_\Delta \approx ROS_{10}$ for every resolution and enough downstream ignitions to fit all 15 ROS values.

The current cell-local representation does not conserve those quantities. Fine meshes fragment houses into multiple emitters, intermediate meshes alter footprints and gaps, and some coarse meshes merge structures or remove firebreaks. The expected diagnostic result is therefore a nonuniform ROS curve, including possible failure to propagate at some resolutions. That observation exposes the defect; against the mesh-independence requirement, its correct overall status is FAIL.

The period of the one-dimensional physical indicator is $a + d = 10 + 10 = 20$ m. The key raster substitutions are therefore:

Δ (m)	mode-resampled consequence	component-area reference
7	one or two selected cells per coordinate	$7^2 = 49$ to $(2 \cdot 7)^2 = 196$ m^2
10	one structure cell followed by one gap cell	$10^2 = 100$ m^2
30	one selected cell per represented structure	$30^2 = 900$ m^2

At $\Delta = 20$ m, each cell contains 10 m of structure and 10 m of gap, so $f_i = 0.5$ and the $f_i \geq 0.5$ rule selects every cell; the represented firebreak vanishes. The same 50-percent selection occurs for the 40-m map. These are reproducible consequences of the generated mode rule, not inferred from the output fire spread.

With $U = 40(0.44704) = 17.8816\ \mathrm{m\,s^{-1}}$, each generated step is

$$\Delta t = \Delta/U; \quad \Delta t_7 = 0.391463, \quad \Delta t_{10} = 0.559234, \quad \Delta t_{30} = 1.677702\ \mathrm{s}.$$

A truly resolution-independent treatment would give all 15 finite fits, $f_{\mathrm{fit}} = 15/15 = 1$, $ROS_\Delta = ROS_{10}$, $CV_R = 0$, and $D_{\max} = 0$. The current native cell treatment is expected instead to produce missing or nonuniform fits because source HRR scales as Δ^2 and the represented geometry changes.

The predeclared mesh-independence requirements permit $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$; exceeding either limit or failing any of the 15 fits yields the expected diagnostic FAIL.

The primary publication provides the independent propagation reference calculation used only for strict stalled-final compatibility: Figure 9 reports zero ROS at 1, 3, 7, 13, 17, 23, 25, and 27 m, and finite propagation at 5, 10, 15, 20, 30, 40, and 50 m [2]. An early stalled-final dump may therefore use the no-propagation exception only for the first set, and only after the meteorology-interval reconstruction and timestep-grid checks also pass. This reference calculation does not replace the current-run metrics or authorize accepting an arbitrary overrun timestamp.

4 Verification metrics and acceptance criteria

Preprocessing passes only when all 15 resolutions are generated, the exact 10-m map has no footprint or gap error, the 7-m footprint range is 49–196 m^2, and the median 30-m footprint is 900 m^2. These checks establish that execution comparison uses the intended resolution-dependent rasterizations.

A complete execution decision additionally requires every input-matched simulation condition and a finite 10-m ROS anchor. Let N_{fit} be the number of resolutions with at least three ignited downwind structure columns. ROS extraction completeness is

$$f_{\text{fit}} = \frac{N_{\text{fit}}}{15},$$

and must equal one. For the finite fitted values, mesh dependence is measured by the coefficient of variation

$$CV_R = \frac{\text{std}_\Delta(ROS_\Delta)}{\text{mean}_\Delta(ROS_\Delta)}$$

and by the largest deviation from the exact-geometry anchor,

$$D_{\max} = \max_\Delta \frac{|ROS_\Delta - ROS_{10}|}{ROS_{10}}.$$

The permissive engineering limits are $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$. Missing fits are handled by f_{fit}; they are not silently discarded and counted as convergence. The overall decision is the conjunction of every geometry, execution, anchor, fit-completeness, and mesh-independence check.

4.1 Justification of metric quantification

The geometry limits are exact consequences of rasterization. At $\Delta = 10$ m, the structure side and gap are both exactly one cell, so the 1% footprint and separation allowances cover only file precision and connected-component bookkeeping. At 7 m, a represented component can contain one to four cells, giving areas from $7^2 = 49$ to $4 \times 7^2 = 196$ m^2. At 30 m, a one-cell represented structure has area $30^2 = 900$ m^2. These checks establish that the intended mesh-dependent maps were generated; they are not claims of geometric accuracy.

For the behavioral requirement, $f_{\text{fit}} = 1$ is non-negotiable because mesh independence cannot be inferred after silently discarding resolutions that fail to propagate. The $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$ limits are intentionally permissive engineering bounds, substantially larger than ordinary front-fit quantization. Even with those allowances, a method with factor-of-two variation or missing fits fails, which is the defect this case is designed to expose [2]. The 10-m reference is the actual ROS_{10} extracted by the same postprocessor, so its plotted reference and ELMFIRE point must coincide by construction; it is not an independent nominal value. Current input identity records and all 15 runs are required to prevent stale or selectively complete outputs from weakening the expected FAIL diagnosis.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Prepared simulation conditions	Generated simulation specification count divided by the 15 requested grid spacings.	100%
10-m footprint error	Maximum relative error of represented component areas from 100 m^2.	≤ 0.01
10-m separation error	Maximum relative gap-width error from 10 m.	≤ 0.01
7-m component areas	Minimum and maximum represented connected-component areas.	49–196 m^2
30-m footprint error	Relative error of median represented footprint from 900 m^2.	≤ 0.01
Execution completeness	Current-input identity record final time of arrival (TOA) output and successful termination.	15 of 15 simulation conditions
10-m ROS anchor	Structure-column fit from the current 10-m TOA raster.	Finite ROS_{10}
ROS extraction completeness	Resolutions with at least three ignited downwind columns.	$f_{\text{fit}} = 1$
ROS mesh-dependence CV	Coefficient of variation over finite fitted ROS values.	$CV_R \leq 0.25$
Maximum anchor deviation	Maximum finite-grid relative deviation from ROS_{10}.	$D_{\max} \leq 0.50$
Overall decision	Every geometry, completeness, and ROS component is conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for a 10 m grid. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

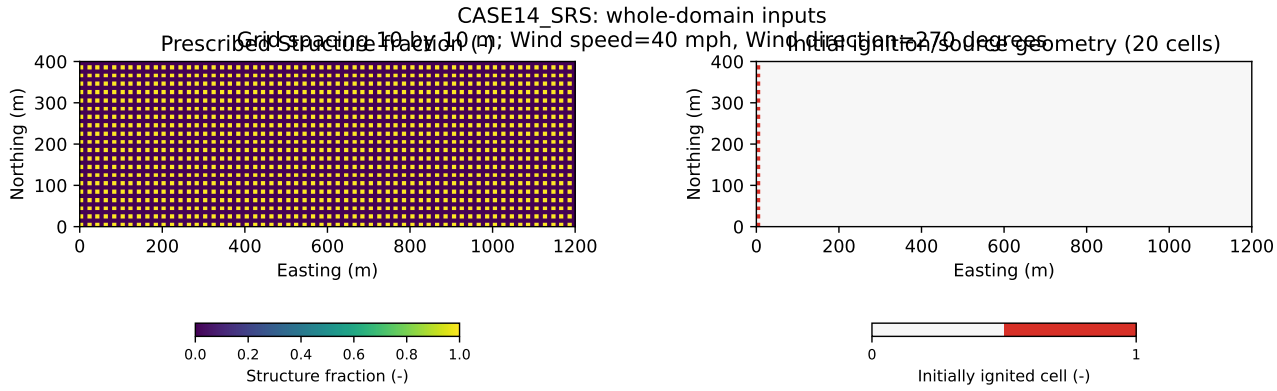


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configuration of the two-dimensional resolution sweep.

Parameter	Configured value	Units	Role
Physical domain	1200×400, flat	m	permits many downwind ignitions
Structure side / gap	10 / 10	m	exact physical scales at the 10-m grid
Initial ignition	complete first structure column	–	uniform upwind source across all structure rows
Grid spacings	1, 3, 5, 7, 10, 13, 15, 17, 20, 23, 25, 27, 30, 40, 50	m	exposes raster and cell-local dependence
Numerical halo	2 on every side	cells	excluded from physical geometry
Wind	40 from 270 degrees	mph	constant west-to-east transport
Time step	Δ/U_{wind}	s	holds wind CFL at one
Design-fire peak	400	kW m^{-2}	structural heat-release rate per unit area (HRRPUA) input
Generation rate	10	$\text{pcs MW}^{-1}\ \text{s}^{-1}$	cell-local building source
Transport / accumulation	empirical / Eulerian	–	current structural spotting path
Ignition delays	42 plus 300	s	small-flame and development stages
Vegetative surface spread	excluded by fuel 93	–	isolates firebrand-driven propagation
Stop time	at least 150000	s	rounded up to an exact timestep multiple
Final output	one TOA per simulation condition	–	source for structure-column ROS fits

The dashed reference line is predicted to intersect the 10-m ELMFIRE point exactly because both are ROS_{10}. A mesh-independent implementation would cluster all other finite ROS points around this line and provide a fit at every resolution. The current independent-cell treatment is instead expected to produce missing fits and large variations, causing the behavioral verification to fail.

5.1 Preprocessing, execution, and postprocessing

input preparation procedure places customizable constants at its beginning, constructs the periodic map analytically, assigns persistent structure IDs, writes co-registered GeoTIFF inputs and case-specific fuel tables, and records a SHA-256 input identity record for every simulation condition. It also writes structure geometry measurements, making the map checks independently auditable.

All 15 prescribed conditions are simulated independently and sequentially. Each result must match the corresponding input configuration before the resolution comparison is evaluated.

result analysis procedure rejects stale or incomplete outputs, reads the final time-of-arrival raster, groups cells by structure, fits ROS, calculates all metrics above, and produces the report figure.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from a 10 m grid. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The 10-m field is the reference-scale spatial result used to interpret mesh-dependent structure spread across the resolution sweep.

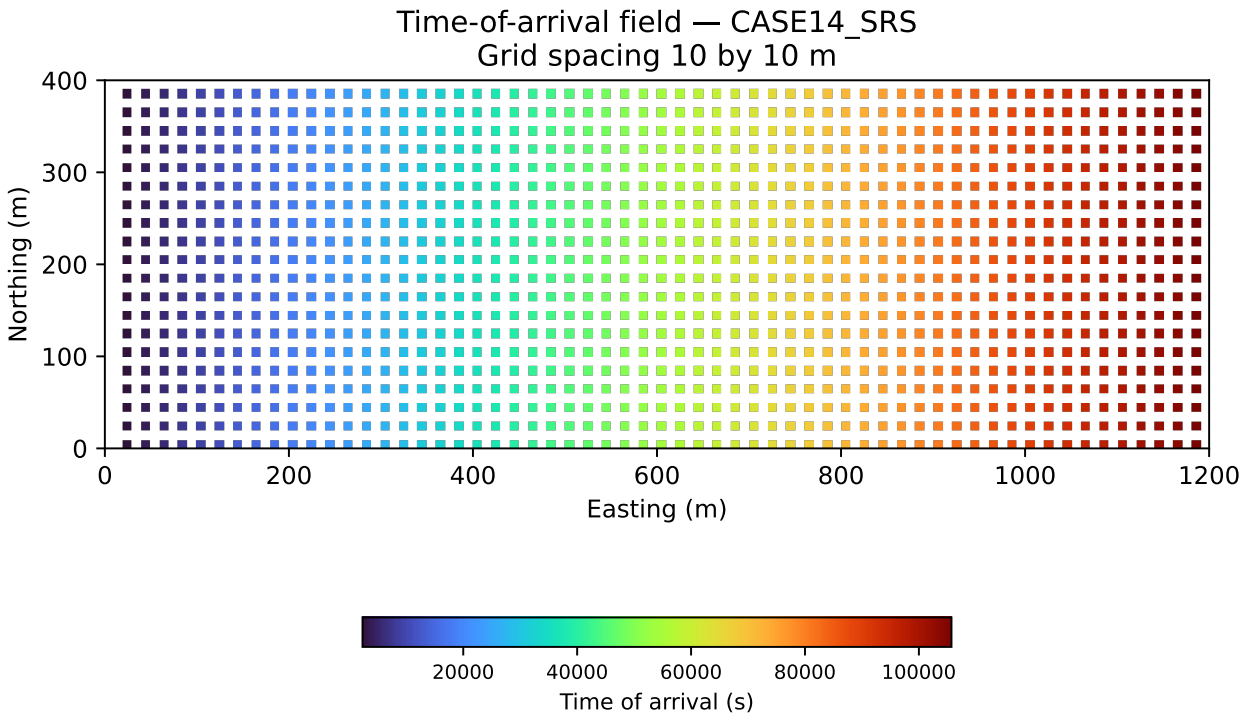


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 15 input-matched ELMFIRE processes reached the source success message. The strict terminal-evidence selector accepts 7 of 15 outputs; process exit alone is not used as evidence. Regular terminal dumps are accepted for 1, 3, 5, 7, 10, 20, 30, and 40 m, and the publication-derived independent prediction of no propagation permits the timestep-aligned stalled-final path for 13, 17, and 25 m. The selected files, configured controls, final times, reconstructed pre-jump times, grid residuals, and rejection reasons are recorded per simulation condition in calculated verification and validation metrics. The overall decision calculated by postprocessing is **NOT EVALUABLE**.

The upper row of Figure 3 shows the actual preprocessed geometries. Its lower panel contains fitted ELMFIRE ROS values. Red crosses at zero identify completed simulations with fewer than three ignited downwind columns, for which ROS is deliberately not fitted. The dashed line is $ROS_{10} = N/A\ \mathrm{m\,s^{-1}}$, exactly equal to the 10-m ELMFIRE point.

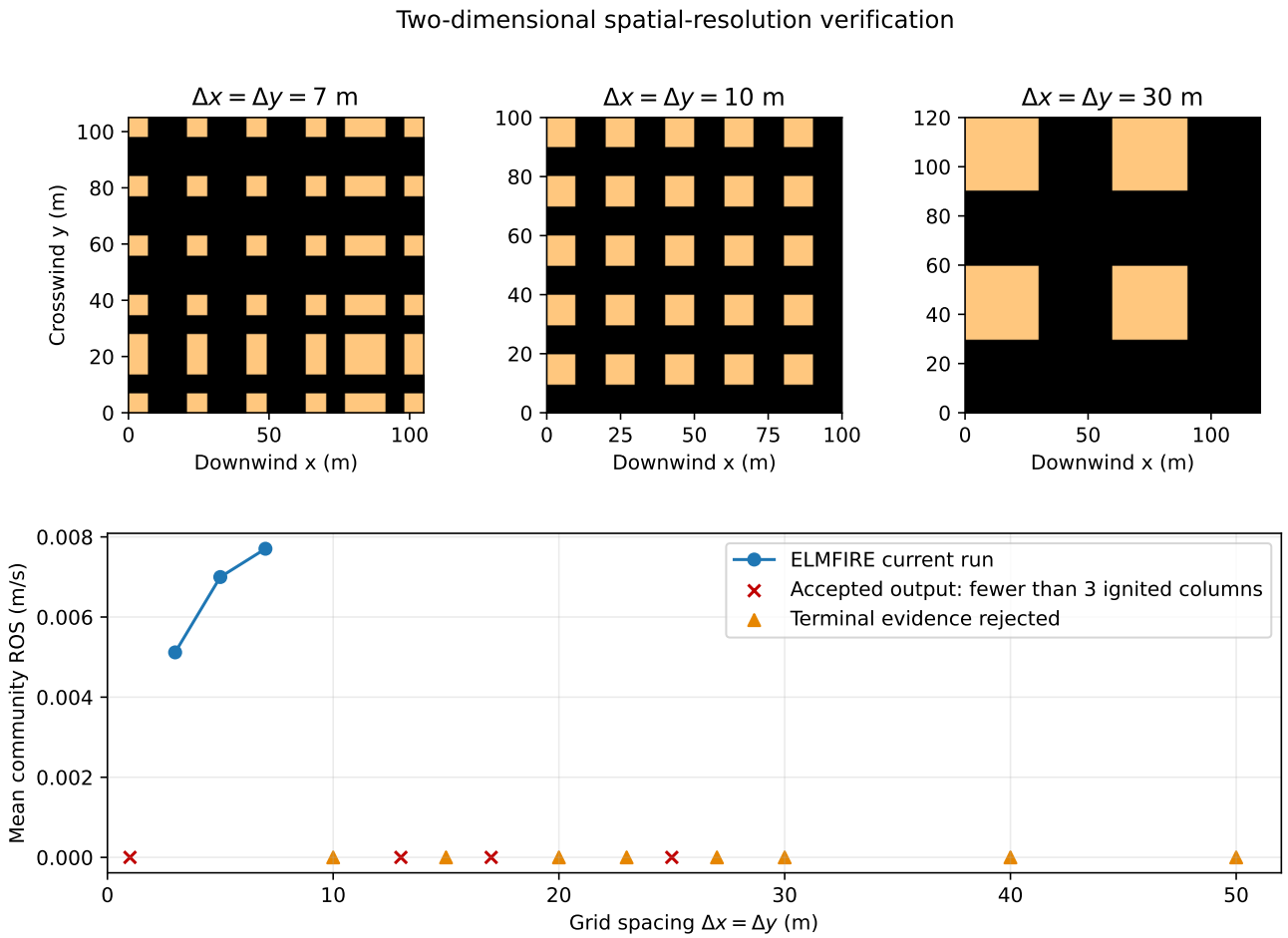


Figure 3: Two-dimensional spatial-resolution diagnostic. Top: mode-resampled WUI geometry at 7, 10, and 30 m. Bottom: mean community ROS from current ELMFIRE TOA outputs; red crosses identify resolutions without enough downwind ignitions for a fit, and the dashed line is the run-derived 10-m anchor. The geometry selection follows the documented comparison configuration [2].

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Prepared resolution simulation conditions	$\geq$ 100%	1	PASS
dx=10 m footprint error	$\leq$ 0.01	0	PASS
dx=10 m separation error	$\leq$ 0.01	0	PASS
dx=7 m component-area range	49 to 196 m2	49 to 196	PASS
dx=30 m median footprint error	$\leq$ 0.01	0	PASS
Current-input identity record run completeness	15 simulation conditions	7/15	NOT EVALUABLE
10-m ELMFIRE ROS anchor	finite fitted ROS	N/A	NOT RUN
ROS-evaluable resolution fraction	$\geq$ 100%	N/A	NOT RUN
ROS mesh-dependence CV	$\leq$ 0.25	N/A	NOT RUN
Maximum ROS deviation from 10-m anchor	$\leq$ 0.50	N/A	NOT RUN

The postprocessor reports **NOT EVALUABLE**. Geometry preparation passes, and all 15 processes executed, but strict terminal evidence is complete for only 7 of 15 simulation conditions. The 10-m anchor, ROS-evaluable fraction, coefficient of variation, and maximum anchor deviation therefore remain `N/A`; calculating them from a selected subset would silently change the predeclared verification population.

8 Technical interpretation

The accepted subset contains finite fitted values of approximately 0.005 12 m s^{-1} (3 m), 0.007 00 m s^{-1} (5 m), 0.007 70 m s^{-1} (7 m), 0.011 23 m s^{-1} (10 m), 0.026 17 m s^{-1} (20 m), 0.016 12 m s^{-1} (30 m), and 0.040 36 m s^{-1} (40 m). These values visibly demonstrate mesh sensitivity, but they are diagnostic observations rather than a complete acceptance calculation.

Four successful processes do not provide admissible terminal evidence. For 15 m, the recorded final time is 152833.592 s and subtracting the configured 3600-s meteorology interval gives 149233.592 s; the timestep-grid residual is 0.302 s. For 23 and 27 m the corresponding reconstructed times are 146507.060 and 148242.909 s, with grid residuals 0.636 and 0.705 s. Those residuals exceed the 0.150-s single-precision-scale tolerance. The 50-m reconstruction is grid-aligned (0.023-s residual), but occurs at 148236.198 s, more than one 2.796-s timestep before the configured stop. The primary publication reports propagation at 15 and 50 m, so neither early stall may be reclassified as independently expected no propagation. Conversely, the publication reports no propagation at 23 and 27 m, but that reference calculation does not waive the timestep-grid requirement.

The current source also produces finite spread at 3 and 7 m while the published Figure 9 places those resolutions at zero, and it stalls at 15 and 50 m where that figure shows finite spread. The resulting **NOT EVALUABLE** status is therefore an implementation/reference and terminal-time discrepancy, not a missing execution and not a reason to weaken the evidence selector. The runs used executable SHA-256 `c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920` built from source commit `a2dfbcdf72209733c000e5d3431e44723e15efea`; the source worktree already contained unrelated local modifications, so the commit and executable digest are both reported for reproducibility.

9 Scope and limitations

This case verifies numerical mesh dependence for one idealized periodic WUI array, one wind, and one structural design fire. It does not validate the physical accuracy of any individual ROS value. The 10-m result is an internal run-derived anchor, not experimental validation data. A future resolution-independent structure implementation should be tested with the same geometry and acceptance criteria; it should provide all 15 ROS fits and reduce both variation metrics below their limits.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

[2] Y. Qin et al., "Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models," *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.